

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459963.7084 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.045
Transit depth (Rp/Rs)^2	2.24 +/- 1.36 [%]
Orbital Inclination (inc)	85.01 +/- 1.75
Airmass coefficient 1 (a1)	0.9947 +/- 0.0093
Airmass coefficient 2 (a2)	0.0049 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#3 - [587, 444]
Optimal Aperture	5.8
Optimal Annulus	12.56
Transit Duration (day)	0.04 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.045 vs 0.131 (deviation 0.42σ)
Duration fit vs published	0.04 d vs 0.1075 d (deviation 1.73σ)
Mid-time O–C	-7.0 min
Depth SNR	3.3
Residual scatter	0.94 %

Light curve

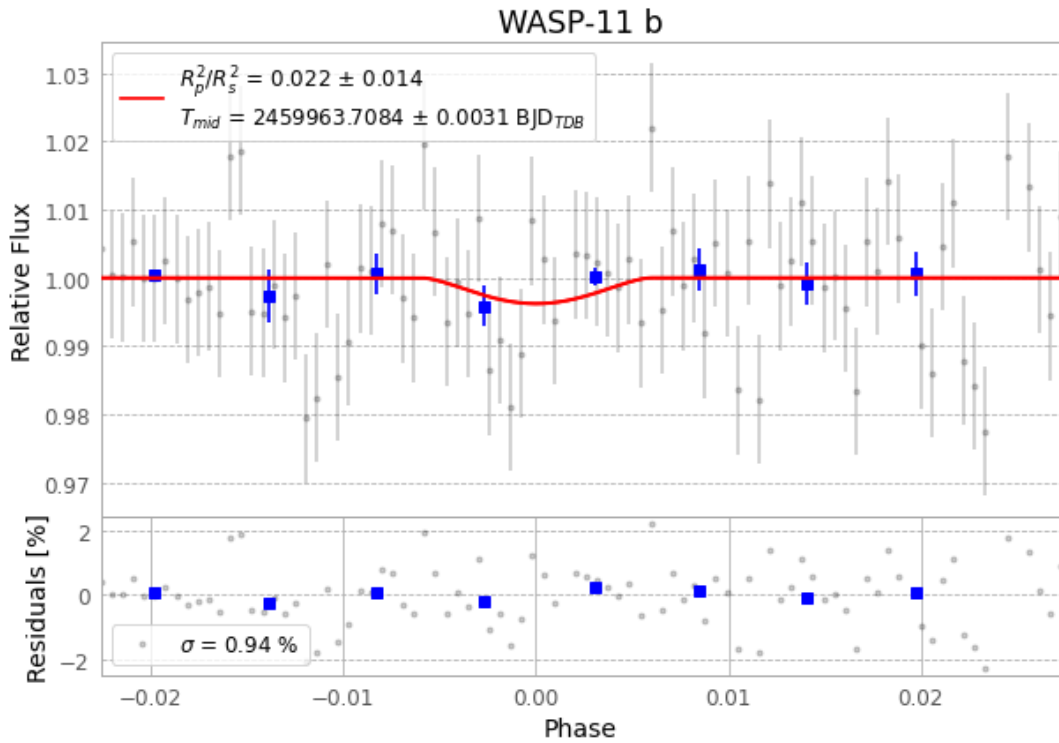


Figure 1 — FinalLightCurve_WASP-11 b_2023-01-18.png (SHA-256 8f55525ed3823a6e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [473, 367], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f7cdc496e68066d6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

90 FITS frames (59 MB), HATP-10230119025416.FITS → HATP-10230119072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-230119.log (94232d224361...), FinalLightCurve_WASP-11 b_2023-01-18.png (8f55525ed382...), FinalLightCurve_WASP-11 b_2023-01-18.csv (2f2645c59589...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:09Z.